# Jungsoo Kim

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## Research Profile

Research on uncertainty and reliability in AI across healthcare, policy evaluation, and evidentiary use, combining conformal prediction, causal representation learning, and image restoration with hands-on experimental design, implementation, and quantitative and qualitative evaluation.

## Education

**Ph.D.** **Seoul National University, Graduate School of Data Science**, Data Science — Seoul, Korea
- Advisor: Sanghack Lee; Causality Lab. — Sept 2024 – present

**M.S.** **Seoul National University, Graduate School of Data Science**, Data Science — Seoul, Korea
- GPA: 4.22/4.5 — Mar 2022 – Aug 2024

**B.P.A. / B.B.** **Yonsei University, Underwood International College**, Science, Technology and Policy; Quantitative Risk Management — Seoul, Korea
- GPA: 4.15/4.5 — Mar 2017 – Feb 2022

**Exchange Student** **National University of Singapore, Yale-NUS College**, Mathematical and Computational Science — Singapore, Sept 2019 – Feb 2020

## Selected Publications

**Decomposing Conformal Uncertainty: Calibration- and Instance-Driven Feature Attribution** — Jan 2026
Accepted as a poster. Equal contribution (co-first authors): Sangyeon Cho, Minyoung Cho, and Jungsoo Kim.
Sangyeon Cho, Minyoung Cho, *Jungsoo Kim*, Sujeong Oh, Sanghack Lee
(Advances in Neural Information Processing Systems (NeurIPS))

**A Tale of Two Uncertainties: Global-Local Attribution for Conformal Prediction** — Jan 2026
Spotlight paper; earlier workshop version of the NeurIPS 2026 paper. Equal contribution (co-first authors): Sangyeon Cho, Minyoung Cho, and Jungsoo Kim.
Sangyeon Cho, Minyoung Cho, *Jungsoo Kim*, Sujeong Oh, Sanghack Lee
openreview.net/forum?id=2alvedJMpp (ICML 2026 Workshop on Epistemic Intelligence in Machine Learning (EIML))

**Instrumental Variable Representation Learning under Confounded Covariates** — Jan 2025
*Jungsoo Kim*, Kwonho Kim, Inwoo Hwang, Sanghack Lee
(NeurIPS 2025 Workshop on CauScien: Uncovering Causality in Science)

**When Mass Atrocities Come to Light in the Artificial Eyes in the Sky: AI and the Use of Satellite Images for International Criminal Accountability**
Under review.
*Jungsoo Kim*, Heejin Kim, Sanghack Lee
(Manuscript under review)

## Selected Research Projects

**Conformal uncertainty and patient-specific retesting**

- Originated the motivating use case and led experimental design: constructed semi-synthetic data from real sepsis data and designed a risk score with patient-specific retesting needs.
- Compared retest-variable selection using total attribution, equal calibration/instance weighting, and greater instance weighting; showed that emphasizing the test instance better prioritized patient-specific retest variables under a limited testing budget in simulation.

**Instrumental variable representation learning under confounded covariates**

- Conceived the research question and designed an architecture to learn IV representations from high-dimensional covariates correlated with unobserved confounders, targeting causal effect estimation under weaker assumptions.
- Implemented the model, ran the experiments, and wrote the manuscript, carrying the project from initial concept through empirical evaluation.

**Reliability of AI-enhanced satellite imagery as evidence**

- Investigated the reliability of AI-enhanced satellite imagery for international criminal accountability; ran model inference across 15 super-resolution setups and conducted quantitative and qualitative analyses.
- Combined quantitative and qualitative evidence of reliability limitations with analysis of international criminal accountability, highlighting risks of the evaluated AI-enhanced imagery as evidence; manuscript under review.

## Research and Industrial Experience

**Ministry of Food and Drug Safety (MFDS) Project**, Project Lead for Causal Inference Section — Mar 2025 – Dec 2025

- Led the causal inference section of the MFDS project, applying conformal prediction to analyses of statin effects on blood glucose.
- Investigated variation in prediction interval widths across individuals; this practical problem motivated subsequent research on feature attribution for conformal uncertainty.

**SK Center for Social Value Enhancement Studies**, Research Assistant — Aug 2021 – Jan 2022

- Analyzed beneficiary financial records totaling KRW 10 billion, estimated the following fiscal year's budget, and built a research database on the Korean social economy.

**Mooncademy (Korean Traditional Music Education Start-up)**, Product Manager — Oct 2020 – Apr 2021

- Planned website information architecture with engineers and designers; launched a gayageum lecture crowdfunding campaign raising KRW 10,491,700.

## Teaching and Service

**Teaching Assistant, Seoul National University**
Causal Inference; Machine Learning and Deep Learning II; Basic Statistics & Mathematics Bootcamp.

**Python Instructor, SNU Language Education Institute Outreach** — 2025-07 to 2025-08
Taught Python and basic algorithms to Malaysian high school students, emphasizing collaborative problem-solving.

## Selected Honors and Writing

**5th POSTECH SF Award, Excellence Prize** — Jan 2025
Authored "Will of Stochastic Ghost," published in Diagonal Argument (Eunhengnamu, 2025); fiction exploring generative AI and evidence.

**Journal of Korea Artificial Intelligence Association, Best Paper** Jan 2024
Submitted a master's project paper on instrumental variables.

**CSES Social Value Research Idea Award, Excellence Prize** Jan 2024
Proposed a study on causal structure analysis for Korean population policy.

**LH COMPAS Land & City Data Competition, Excellence Prize** Jan 2024
Identified medically underserved areas in Asan through geospatial analysis.

**EdTech Data Analysis Hackathon, Grand Prize** Jan 2023
Built a dropout-prediction model using survival analysis.

**Public BigData Analysis Hackathon, Excellence Prize** Jan 2020
Developed a traffic-accident risk model with XGBoost and interpreted drivers using SHAP.

## Skills and Languages

**Programming:** Python, R, C++, C, SQL

**Research Methods:** Conformal prediction and uncertainty quantification; instrumental variables and causal representation learning; experimental design; quantitative and qualitative model evaluation

**ML and Geospatial Tools:** PyTorch; super-resolution and diffusion model workflows (SRCNN, SRGAN, StableSR, Stable Diffusion); GeoPandas, QGIS

**Data Systems:** Neo4j, BigQuery; Kafka (event streaming)

**Languages:** Korean, English (TOEFL 100; OPIc AL)